## 1. CONCAT Function

**Definition:** The CONCAT function combines text from multiple cells into a single cell without automatically adding any separator between the text.

**Syntax:**
=CONCAT(text1, [text2], ...)

**Example:**

A2 = Shagun
B2 = Sharma

Formula: =CONCAT(A2,B2)

Output: ShagunSharma

To add a space between the words:

Formula: =CONCAT(A2," ",B2)

Output: Shagun Sharma

## 2. TEXTJOIN Function

**Definition:** The TEXTJOIN function combines text from multiple cells into a single cell using a specified delimiter. It can also ignore empty cells.

**Syntax:**
=TEXTJOIN(delimiter, ignore_empty, text1, ...)

**Example:**

A2 = Apple
B2 = Banana
C2 = Mango

Formula: =TEXTJOIN(", ",TRUE,A2:C2)

Output: Apple, Banana, Mango

Here:

- ", " specifies a comma followed by a space as the delimiter.
- TRUE tells Excel to ignore empty cells.
- A2:C2 is the range of cells to combine.

## 3. Delimiter

**Definition:** A delimiter is a character or symbol used to separate different pieces of text.

**Examples:**

- Comma (,): Apple,Banana,Mango
- Space ( ): Apple Banana Mango
- Hyphen (-): Apple-Banana-Mango
- Forward slash (/): Apple/Banana/Mango
- Semicolon (;): Apple;Banana;Mango

In Excel, a delimiter specifies what should appear between text values when they are joined.

**4. Difference Between CONCAT and TEXTJOIN**

**CONCAT:**

- Combines text from multiple cells.
- Does not automatically insert a separator between text values.
- Useful when text needs to be combined directly.

**TEXTJOIN:**

- Combines text from multiple cells using a specified delimiter.
- Can ignore empty cells.
- Useful when text needs to be separated by commas, spaces, hyphens, or other characters.

**5. TEXTSPLIT Function in Excel**

**Definition:** The TEXTSPLIT function divides text from a single cell into multiple cells based on a specified delimiter.

**Syntax:**
=TEXTSPLIT(text, col_delimiter, [row_delimiter])

**How to Use:**

- To split text into columns: =TEXTSPLIT(A2,",")
- To split text into rows: =TEXTSPLIT(A2,,",")

**Important Points:**

- TEXTSPLIT separates text into multiple cells.
- It can split text horizontally into columns or vertically into rows.
- It uses delimiters such as commas, spaces, hyphens, or other characters.

- It is useful for separating full names, dates, and other combined text.
- Unlike CONCAT and TEXTJOIN, which combine text, TEXTSPLIT separates text.

**LEFT Function**

**Definition:** The LEFT function extracts a specified number of characters from the beginning (left side) of a text string.

**Syntax:**
=LEFT(text, [num_chars])

**Example:**
Formula: =LEFT("Excel",2)
Output: Ex

**RIGHT Function**

**Definition:** The RIGHT function extracts a specified number of characters from the end (right side) of a text string.

**Syntax:**
=RIGHT(text, [num_chars])

**Example:**
Formula: =RIGHT("Excel",2)
Output: el

**Important Points**

- Both functions extract characters from a text string.
- The number of characters to extract is specified by num_chars.
- If num_chars is omitted, Excel extracts one character by default.
- If num_chars is greater than the length of the text, the entire text is returned.
- These functions are useful for extracting initials, area codes, product codes, and other parts of text.

**FIND Function**

**Definition:** The FIND function returns the position of a specific character or text within another text string.

**Syntax:**
=FIND(find_text, within_text, [start_num])

**Example:**
Formula: =FIND("c","Excel")
Output: 3

**Important Points:**

- FIND is case-sensitive.
- It returns the position of the first occurrence of the specified text.
- The position count starts from 1.
- If the specified text is not found, Excel returns a #VALUE! error.

## MID Function

**Definition:** The MID function extracts a specified number of characters from the middle of a text string, starting at a given position.

**Syntax:**
=MID(text, start_num, num_chars)

**Example:**
Formula: =MID("Microsoft",3,4)
Output: cros

**Important Points:**

- text specifies the text string.
- start_num specifies the position from which extraction begins.
- num_chars specifies the number of characters to extract.
- The position count starts from 1.
- MID is useful for extracting specific parts of names, codes, and other text strings.

## Date and Time Functions

### 1. DAY Function

**Definition:** The DAY function extracts the day of the month from a given date as a number between 1 and 31.

**Syntax:** =DAY(serial_number)

**Example:** =DAY("15-Aug-2025")

**Output:** 15

### 2. MONTH Function

**Definition:** The MONTH function extracts the month from a given date as a number between 1 and 12.

**Syntax:** =MONTH(serial_number)

**Example:** =MONTH("15-Aug-2025")

**Output:** 8

**3. YEAR Function**

**Definition:** The YEAR function extracts the year from a given date.

**Syntax:** =YEAR(serial_number)

**Example:** =YEAR("15-Aug-2025")

**Output:** 2025

**4. DATE Function**

**Definition:** The DATE function creates a valid date using the year, month, and day values.

**Syntax:** =DATE(year, month, day)

**Example:** =DATE(2025,8,15)

**Output:** 15-Aug-2025 (depending on the cell's date format)

**5. TODAY Function**

**Definition:** The TODAY function returns the current date. The date updates automatically when Excel recalculates.

**Syntax:** =TODAY()

**Example:** =TODAY()

**Output:** Current date

**6. NOW Function**

**Definition:** The NOW function returns the current date and time. The result updates automatically when Excel recalculates.

**Syntax:** =NOW()

**Example:** =NOW()

**Output:** Current date and time

**7. HOUR Function**

**Definition:** The HOUR function extracts the hour from a time value as a number between 0 and 23.

**Syntax:** =HOUR(serial_number)

**Example:** =HOUR("3:45 PM")

**Output:** 15

**8. MINUTE Function**

**Definition:** The MINUTE function extracts the minutes from a time value as a number between 0 and 59.

**Syntax:** =MINUTE(serial_number)

**Example:** =MINUTE("3:45 PM")

**Output:** 45

**9. SECOND Function**

**Definition:** The SECOND function extracts the seconds from a time value as a number between 0 and 59.

**Syntax:** =SECOND(serial_number)

**Example:** =SECOND("3:45:30 PM")

**Output:** 30

**10. WEEKDAY Function**

**Definition:** The WEEKDAY function returns a number representing the day of the week for a given date.

**Syntax:** =WEEKDAY(serial_number, [return_type])

**Example:** =WEEKDAY(DATE(2025,8,15),2)

**Output:** 5 (Friday)

Here, 2 means Monday = 1, Tuesday = 2, and so on, with Sunday = 7.

**11. DAYS Function**

**Definition:** The DAYS function calculates the number of days between two dates.

**Syntax:** =DAYS(end_date, start_date)

**Example:** =DAYS(DATE(2025,8,15),DATE(2025,8,1))

**Output:** 14

**12. DATEDIF Function**

**Definition:** The DATEDIF function calculates the difference between two dates in years, months, or days.

**Syntax:** =DATEDIF(start_date, end_date, unit)

**Example:** =DATEDIF(DATE(2020,1,1),DATE(2025,1,1),"Y")

**Output:** 5

Common units:

- "Y" – Difference in complete years
- "M" – Difference in complete months
- "D" – Difference in days

**Important Points**

- Excel stores dates as serial numbers, allowing calculations using dates.
- Date and time results can be displayed differently by changing cell formatting.
- TODAY() returns the current date, while NOW() returns the current date and time.
- DAY, MONTH, and YEAR extract individual components of a date.
- HOUR, MINUTE, and SECOND extract individual components of a time.

**1. HOUR Function**

**Definition:** The HOUR function extracts the hour from a time value as a number between 0 and 23.

**Syntax:** =HOUR(serial_number)

**Example:** =HOUR("3:45 PM")

**Output:** 15

**2. MINUTE Function**

**Definition:** The MINUTE function extracts the minutes from a time value as a number between 0 and 59.

**Syntax:** =MINUTE(serial_number)

**Example:** =MINUTE("3:45 PM")

**Output:** 45

**3. SECOND Function**

**Definition:** The SECOND function extracts the seconds from a time value as a number between 0 and 59.

**Syntax:** =SECOND(serial_number)

**Example:** =SECOND("3:45:30 PM")

**Output:** 30

**Important Points**

- HOUR extracts the hour component of a time.
- MINUTE extracts the minute component of a time.
- SECOND extracts the second component of a time.
- Excel stores time as a fraction of a day.
- These functions return numbers, not formatted time values.